

Suppose now that $h = 120$. The parametric form for the solution the above vector equation is

$$\begin{aligned}x &= -.2z \\ y &= -.8z.\end{aligned}$$

Taking $z = 1$ gives the linear combination

$$-.2 \begin{pmatrix} 180 \\ 50 \\ 200 \end{pmatrix} - .8 \begin{pmatrix} 100 \\ 150 \\ 100 \end{pmatrix} + \begin{pmatrix} 116 \\ 130 \\ 120 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

In terms of colors:

$$\begin{pmatrix} 116 \\ 130 \\ 120 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 180 \\ 50 \\ 200 \end{pmatrix} + \frac{4}{5} \begin{pmatrix} 100 \\ 150 \\ 100 \end{pmatrix} = \begin{pmatrix} 36 \\ 10 \\ 40 \end{pmatrix} + \begin{pmatrix} 80 \\ 120 \\ 80 \end{pmatrix}$$











3. Which of the following must be true for any set of seven vectors in $\mathbf{R}^5$? Answer “yes”, “no”, or “maybe” in each case.
- The vectors span $\mathbf{R}^5$.
 - The vectors are linearly dependent.
 - At least one of the vectors is in the span of the other six vectors.
 - If we put the seven vectors as the columns of a matrix A , then the matrix equation $Ax = 0$ must have infinitely many solutions.
 - Suppose we put the seven vectors as the columns of a matrix A . Then for each b in $\mathbf{R}^5$, the matrix equation $Ax = b$ must be consistent.
 - If every vector b in $\mathbf{R}^5$ can be written as a linear combination of our seven vectors, then in fact every b in $\mathbf{R}^5$ can be written in *infinitely many* different ways as a linear combination of our seven vectors.

Solution.

- Maybe.
- Yes.
- Yes.
- Yes.
- Maybe.
- Yes. By assumption, the matrix A whose columns are our seven vectors has $\mathbf{R}^5$ as its column span, so A will have a pivot in every row. Therefore, A will have 5 pivot columns, so it will have 2 columns without pivots. This means

that $Ax = b$ will be consistent no matter what b is, and there will be two free variables.

4. Suppose A is a 2×3 matrix and the solution set to $Ax = 0$ is $\text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$. Must it be true that the equation $Ax = b$ is consistent for each b in $\mathbf{R}^2$?

Solution.

Yes. The matrix equation $Ax = 0$ has one free variable because its solution set is a line. Since A is a 2×3 matrix, this means A has two pivots, so it has a pivot in every row. Therefore, the columns of A span $\mathbf{R}^2$, so the matrix equation $Ax = b$ is consistent for each b in $\mathbf{R}^2$.

5. Write vectors u , v , and w in $\mathbf{R}^4$ so that $\{u, v, w\}$ is linearly dependent, but u is not in $\text{Span}\{v, w\}$.

Supplemental problems: §§2.6, 2.7, 2.9

1. Circle **TRUE** if the statement is always true, and circle **FALSE** otherwise.

a) If A is a 3×100 matrix of rank 2, then $\dim(\text{Nul } A) = 97$.

TRUE FALSE

b) If A is an $m \times n$ matrix and $Ax = 0$ has only the trivial solution, then the columns of A form a basis for $\mathbf{R}^m$.

TRUE FALSE

c) The set $V = \left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \text{ in } \mathbf{R}^4 \mid x - 4z = 0 \right\}$ is a subspace of $\mathbf{R}^4$.

TRUE FALSE

Solution.

a) False. By the Rank Theorem, $\text{rank}(A) + \dim(\text{Nul } A) = 100$, so $\dim(\text{Nul } A) = 98$.

b) False. For example, $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ has only the trivial solution for $Ax = 0$, but its column space is a 2-dimensional subspace of $\mathbf{R}^3$.

c) True. V is $\text{Nul}(A)$ for the 1×4 matrix A below, and therefore is automatically a subspace of $\mathbf{R}^4$:

$$A = \begin{pmatrix} 1 & 0 & -4 & 0 \end{pmatrix}.$$

Alternatively, we could verify the subspace properties directly if we wished, but this is much more work!

(1) The zero vector is in V , since $0 - 4(0)0 = 0$.

(2) Let $u = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ w_1 \end{pmatrix}$ and $v = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \\ w_2 \end{pmatrix}$ be in V , so $x_1 - 4z_1 = 0$ and $x_2 - 4z_2 = 0$.

We compute

$$u + v = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \\ w_1 + w_2 \end{pmatrix}.$$

Is $(x_1 + x_2) - 4(z_1 + z_2) = 0$? Yes, since

$$(x_1 + x_2) - 4(z_1 + z_2) = (x_1 - 4z_1) + (x_2 - 4z_2) = 0 + 0 = 0.$$

(3) If $u = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$ is in V then so is cu for any scalar c :

$$cu = \begin{pmatrix} cx \\ cy \\ cz \\ cw \end{pmatrix} \quad \text{and} \quad cx - 4cz = c(x - 4z) = c(0) = 0.$$

2. Write a matrix A so that $\text{Col}A = \text{Span} \left\{ \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \right\}$ and $\text{Nul}A$ is the xz -plane.

Solution.

Many examples are possible. We'd like to design an A with the prescribed column span, so that $(A \mid 0)$ will have free variables x_1 and x_3 . One way to do this is simply to leave the x_1 and x_3 columns blank, and make the second column $\begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix}$. This guarantees that A destroys the xz -plane and has the column span required.

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

An alternative method for finding the same matrix: Write $A = (v_1 \ v_2 \ v_3)$. We want the column span to be the span of $\begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix}$ and we want

$$A \begin{pmatrix} x \\ 0 \\ z \end{pmatrix} = (v_1 \ v_2 \ v_3) \begin{pmatrix} x \\ 0 \\ z \end{pmatrix} = xv_1 + zv_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{for all } x \text{ and } z.$$

One way to do this is choose $v_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, and $v_2 = \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix}$.

3. Circle **T** if the statement is always true, and circle **F** otherwise. You do not need to explain your answer.
- a) If $\{v_1, v_2, v_3, v_4\}$ is a basis for a subspace V of $\mathbf{R}^n$, then $\{v_1, v_2, v_3\}$ is a linearly independent set.
 - b) The solution set of a consistent matrix equation $Ax = b$ is a subspace.
 - c) A translate of a span is a subspace.

Solution.

- a) True. If $\{v_1, v_2, v_3\}$ is linearly dependent then $\{v_1, v_2, v_3, v_4\}$ is automatically linearly dependent, which is impossible since $\{v_1, v_2, v_3, v_4\}$ is a basis for a subspace.
- b) False. this is true if and only if $b = 0$, i.e., the equation is *homogeneous*, in which case the solution set is the null space of A .
- c) False. A subspace must contain 0.
4. True or false (justify your answer). Answer true if the statement is *always* true. Otherwise, answer false.
- a) There exists a 3×5 matrix with rank 4.
- b) If A is an 9×4 matrix with a pivot in each column, then
- $$\text{Nul}A = \{0\}.$$
- c) There exists a 4×7 matrix A such that nullity $A = 5$.
- d) If $\{v_1, v_2, \dots, v_n\}$ is a basis for $\mathbf{R}^4$, then $n = 4$.

Solution.

- a) False. The rank is the dimension of the column space, which is a subspace of $\mathbf{R}^3$, hence has dimension at most 3.
- b) True.
- c) True. For instance,

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

- d) True. Any basis of $\mathbf{R}^4$ has 4 vectors.

5. Find bases for the column space and the null space of

$$A = \begin{pmatrix} 0 & 1 & -3 & 1 & 0 \\ 1 & -1 & 8 & -7 & 1 \\ -1 & -2 & 1 & 4 & -1 \end{pmatrix}.$$

Solution.

The RREF of $(A \mid 0)$ is

$$\left(\begin{array}{ccccc|c} 1 & 0 & 5 & -6 & 1 & 0 \\ 0 & 1 & -3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

so x_3, x_4, x_5 are free, and

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -5x_3 + 6x_4 - x_5 \\ 3x_3 - x_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = x_3 \begin{pmatrix} -5 \\ 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 6 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore, a basis for $\text{Nul } A$ is $\left\{ \begin{pmatrix} -5 \\ 3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$

To find a basis for $\text{Col } A$, we use the pivot columns as they were written in the *original* matrix A , not its RREF. These are the first two columns:

$$\left\{ \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right\}.$$

6. Find a basis for the subspace V of $\mathbf{R}^4$ given by

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} \text{ in } \mathbf{R}^4 \mid x + 2y - 3z + w = 0 \right\}.$$

Solution.

V is $\text{Nul } A$ for the 1×4 matrix $A = \begin{pmatrix} 1 & 2 & -3 & 1 \end{pmatrix}$. The augmented matrix $(A \mid 0) = \begin{pmatrix} 1 & 2 & -3 & 1 & 0 \end{pmatrix}$ gives $x = -2y + 3z - w$ where y, z, w are free variables. The parametric vector form for the solution set to $Ax = 0$ is

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -2y + 3z - w \\ y \\ z \\ w \end{pmatrix} = y \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore, a basis for V is

$$\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

7. a) True or false: If A is an $m \times n$ matrix and $\text{Nul}(A) = \mathbf{R}^n$, then $\text{Col}(A) = \{0\}$.
 b) Give an example of 2×2 matrix whose column space is the same as its null space.

- c) True or false: For some m , we can find an $m \times 10$ matrix A whose column span has dimension 4 and whose solution set for $Ax = 0$ has dimension 5.

Solution.

- a) If $\text{Nul}(A) = \mathbf{R}^n$ then $Ax = 0$ for all x in $\mathbf{R}^n$, so the only element in $\text{Col}(A)$ is $\{0\}$.
Alternatively, the rank theorem says

$$\dim(\text{Col } A) + \dim(\text{Nul } A) = n \implies \dim(\text{Col } A) + n = n \implies \dim(\text{Col } A) = 0 \implies \text{Col } A = \{0\}.$$

- b) Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Its null space and column space are $\text{Span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$.

- c) False. The rank theorem says that the dimensions of the column space ($\text{Col } A$) and homogeneous solution space ($\text{Nul } A$) add to 10, no matter what m is.

8. Suppose V is a 3-dimensional subspace of $\mathbf{R}^5$ containing $\begin{pmatrix} 1 \\ -4 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ -3 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 9 \\ 8 \\ 1 \\ 0 \\ 1 \end{pmatrix}$.

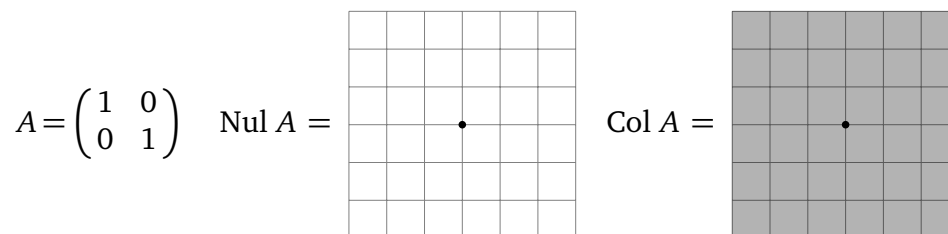
$$\text{Is } \left\{ \begin{pmatrix} 1 \\ -4 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 9 \\ 8 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ a basis for } V? \text{ Justify your answer.}$$

Solution.

Yes. The Basis Theorem says that since we know $\dim(V) = 3$, our three vectors will form a basis for V if and only if they are linearly independent.

Call the vectors v_1, v_2, v_3 . It is very little work to show that the matrix $A = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ has a pivot in every column, so the vectors are linearly independent.

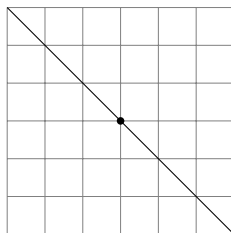
9. a) Write a 2×2 matrix A with **rank** 2, and draw pictures of $\text{Nul } A$ and $\text{Col } A$.



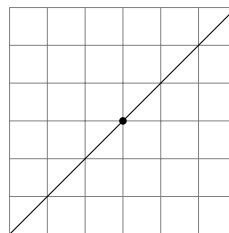
- b) Write a 2×2 matrix B with **rank** 1, and draw pictures of $\text{Nul } B$ and $\text{Col } B$.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Nul $B =$



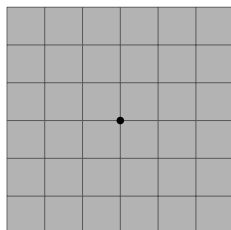
Col $B =$



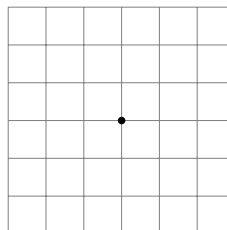
c) Write a 2×2 matrix C with **rank** 0, and draw pictures of Nul C and Col C .

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Nul $C =$



Col $C =$



(In the grids, the dot is the origin.)

Supplemental problems: §3.1

1. Review from 2.6-2.9. Fill in the blanks: If A is a 7×6 matrix and the solution set for $Ax = 0$ is a plane, then the column space of A is a 4-dimensional subspace of $\mathbb{R}^{\boxed{7}}$.

Reason: $\text{rank}(A) + \text{nullity}(A) = 6$ $\text{rank}(A) + 2 = 6$ $\text{rank}(A) = 4$

2. Review from 2.6-2.9: Consider the matrix A below and its RREF:

$$A = \begin{pmatrix} 1 & 2 & -1 & -1 \\ -2 & -4 & -6 & 2 \\ 1 & 2 & -5 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

- a) Write a basis for $\text{Col } A$.

The pivot columns (1 and 3) form a basis for $\text{Col}(A)$, but really column 3 and any other column will work.

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -6 \\ -5 \end{pmatrix} \right\}.$$

- b) Find a basis for $\text{Nul } A$. From the RREF of A , we see the solution set is

$$x_1 + 2x_2 - x_4 = 0, \quad x_3 = 0,$$

so $x_1 = -2x_2 + x_4$, x_2 and x_4 are free, and $x_3 = 0$.

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2x_2 + x_4 \\ x_2 \\ 0 \\ x_4 \end{pmatrix} = x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad \text{A basis is } \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

- c) Is there a matrix B so that $\text{Col}(B) = \text{Nul}(A)$? If yes, write such a B . If not, justify why no such matrix B exists.

Yes. Just take the columns of B to be a set whose span is $\text{Nul } A$, for example

$$B = \begin{pmatrix} -2 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

3. Suppose T is a matrix transformation and the range of T is the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x - 3y + 4z = 0 \right\}$$

of $\mathbf{R}^3$, which contains the vectors $v_1 = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix}$. Is $\{v_1, v_2\}$ a basis for the range of T ?

Solution.

Yes. We know that V is a 2-dimensional subspace of $\mathbf{R}^3$ since $V = \text{Nul} \begin{pmatrix} 1 & -3 & 4 \end{pmatrix}$ which corresponds to a homogeneous system with two free variables. Since $\{v_1, v_2\}$ is clearly a linearly independent set in V and $\dim(V) = 2$, it forms a basis for V by the Basis Theorem.

4. True or false. If the statement is *always* true, answer TRUE. Otherwise, circle FALSE.

a) The matrix transformation $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ performs reflection across the x -axis in $\mathbf{R}^2$. TRUE FALSE (T reflects across the y -axis then projects onto the x -axis)

b) The matrix transformation $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ performs rotation counter-clockwise by 90° in $\mathbf{R}^2$. TRUE FALSE (T rotates clockwise 90°)

5. Let T be the matrix transformation $T(x) = Ax$, where $A = \begin{pmatrix} 1 & 1 & 2 & 1 \\ -1 & 0 & -1 & -2 \\ 2 & 2 & 4 & 2 \end{pmatrix}$.

What is the domain of T ? What is its codomain? Find a basis for the range of T and a basis for the kernel of T (the kernel of T is the set of all vectors satisfying $T(x) = 0$).

Solution: The domain of T is $\mathbf{R}^4$ and the codomain of T is $\mathbf{R}^3$. The range of T is $\text{Col } A$ and the kernel of T is $\text{Nul } A$. We row-reduce $(A \mid 0)$:

$$\left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 0 \\ -1 & 0 & -1 & -2 & 0 \\ 2 & 2 & 4 & 2 & 0 \end{array} \right) \xrightarrow[R_3=R_3-2R_1]{R_2=R_2+R_1} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_1=R_1-R_2} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

We see x_3 and x_4 are free, and $x_1 = -x_3 - 2x_4$ and $x_2 = -x_3 + x_4$. The parametric vector form for elements of $\text{Nul } A$ is:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -x_3 - 2x_4 \\ -x_3 + x_4 \\ x_3 \\ x_4 \end{pmatrix} = x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix}. \text{ A basis for } \text{kernel}(T) \text{ is } \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$